11. Create a DataFrame of ten rows and four columns with random values. Convert some values to NaN values. Highlight the NaN values.

AIM

To create a Pandas DataFrame with random values, replace some values with NaN, and highlight the NaN values.

INPUT

- 10 rows
- 4 columns
- Random values
- Some values converted to NaN

CODE

exp11.py > ...

```python
import pandas as pd
import numpy as np
import webbrowser
import os
df = pd.DataFrame(
    np.random.randint(1, 100, size=(10, 4)),
    columns=['A', 'B', 'C', 'D']
)
df.iloc[2, 1] = np.nan
df.iloc[5, 3] = np.nan
df.iloc[7, 0] = np.nan
def highlight_nan(val):
    if pd.isna(val):
        return "background-color: yellow; color: black;"
    return ""
print("Original DataFrame:")
print(df)
styled_df = df.style.map(highlight_nan)
file_name = "highlight_nan.html"
styled_df.to_html(file_name)
print(f"\nHTML file created successfully: {file_name}")
webbrowser.open('file://' + os.path.realpath(file_name))
```

OUTPUT

```
Original DataFrame:
      A     B   C     D
0  49.0   1.0  27   3.0
1  43.0  89.0  62   3.0
2  67.0   NaN  25  62.0
3   7.0  95.0  69   1.0
4  50.0  19.0  82  98.0
5  62.0  38.0  97   NaN
6  60.0  86.0  56   6.0
7   NaN  10.0  81  88.0
8  16.0  54.0  60  73.0
9   1.0  82.0  76  44.0
```

	A	B	C	D
0	49.000000	1.000000	27	3.000000
1	43.000000	89.000000	62	3.000000
2	67.000000	nan	25	62.000000
3	7.000000	95.000000	69	1.000000
4	50.000000	19.000000	82	98.000000
5	62.000000	38.000000	97	nan
6	60.000000	86.000000	56	6.000000
7	nan	10.000000	81	88.000000
8	16.000000	54.000000	60	73.000000
9	1.000000	82.000000	76	44.000000

RESULT

Thus, a DataFrame was created and NaN values were successfully highlighted.

---

12. Set DataFrame background color to black and font color to yellow

AIM

To create a DataFrame and set its background color to black and text color to yellow.

INPUT

Random DataFrame (10 × 4)

CODE

exp12.py > ...

```python
import pandas as pd
import numpy as np
import webbrowser
import os
df = pd.DataFrame(
    np.random.randint(1, 100, size=(10, 4)),
    columns=['A', 'B', 'C', 'D']
)
print("Original DataFrame:")
print(df)
# Set background color to black and font color to yellow
styled_df = df.style.set_properties(**{
    'background-color': 'black',
    'color': 'yellow'
})
# Save as HTML
file_name = "black_yellow_dataframe.html"
styled_df.to_html(file_name)
print(f"\nHTML file created successfully: {file_name}")
# Open automatically in the default web browser
webbrowser.open('file://' + os.path.realpath(file_name))
```

OUTPUT

```
Original DataFrame:
    A   B   C   D
0  88  34  30  72
1  71  15  79   4
2  53  14  50  66
3  15  70  84  77
4  95  51  63  24
5  86  82  13  53
6  75  42  58  35
7  92  98  27  24
8  67  72   2  81
9  37  19   1  18
```

	A	B	C	D
0	88	34	30	72
1	71	15	79	4
2	53	14	50	66
3	15	70	84	77
4	95	51	63	24
5	86	82	13	53
6	75	42	58	35
7	92	98	27	24
8	67	72	2	81
9	37	19	1	18

RESULT

Thus, the DataFrame background color was changed to black and the font color to yellow.